Project Design Phase-II
Technology Stack (Architecture & Stack)

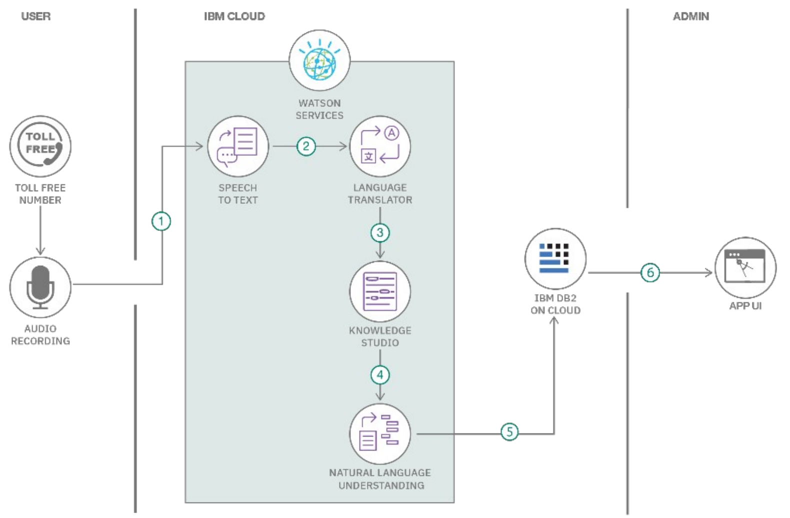
Date	18 July 2026
Team ID	
Project Name	HouseHunt – Premium House Rental Marketplace
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Order processing during pandemics for offline mode

Reference: https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/



Guidelines:

- Include all the processes (As an application logic / Technology Block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third party API's etc.)
- Indicate Data Storage components / services
- Indicate interface to machine learning models (if applicable)

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1. 1	User Interface	How the user interacts with the application (Web UI)	React.js (Vite), Tailwind CSS, Material UI, Bootstrap
2. 2	Application Logic-1	Core request handling and business logic	Node.js, Express.js
3. 3	Application Logic-2	Authentication & role-based authorization logic	JWT, bcryptjs, role-guard middleware
4. 4	Application Logic-3	Booking, negotiation and wallet (GST / commission) processing logic	Express controllers & services
5. 5	Database	Stores users, properties, bookings, wallet, complaint and CMS data	MongoDB (Mongoose ODM)
6. 6	Cloud Database	Managed cloud database cluster	MongoDB Atlas
7. 7	File Storage	Property images and chat attachments	Cloud object storage / static asset hosting
8. 8	External API-1	Live location previews and GPS-based property search	Google Maps API
9. 9	External API-2	Not used in the current scope	—
10. 10	Machine Learning Model	AI-assisted listing description validation and spam scoring	AI-assisted moderation service
11. 11	Infrastructure (Server / Cloud)	Application deployment — client as static site, server as serverless API	Vercel

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1. 1	Open-Source Frameworks	React.js, Express.js, Node.js, Mongoose, Tailwind CSS, Bootstrap	MERN Stack (MongoDB, Express.js, React.js, Node.js), Vercel
2. 2	Security Implementations	JWT-based authentication, bcryptjs password hashing, role-guard middleware, protected API routes	MERN Stack (MongoDB, Express.js, React.js, Node.js), Vercel
3. 3	Scalable Architecture	Three-tier architecture separating presentation, application and data layers with modular controllers/services	MERN Stack (MongoDB, Express.js, React.js, Node.js), Vercel
4. 4	Availability	Deployed on Vercel (client as static site, server as serverless API) with a managed MongoDB cluster	MERN Stack (MongoDB, Express.js, React.js, Node.js), Vercel
5. 5	Performance	Vite fast dev/build pipeline, Axios API layer, indexed Mongoose queries for search and dashboards	MERN Stack (MongoDB, Express.js, React.js, Node.js), Vercel

References:

https://c4model.com/

https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/

https://www.ibm.com/cloud/architecture

https://aws.amazon.com/architecture

https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d